

Syllabus

UG0802 -ZOO-75T-301- Animal Physiology & Biochemistry

UG0802 -ZOO-75P-302- Practicals based on Animal Physiology & Biochemistry

V Semester-Zoology

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
V	ZOO-75T-301 ZOO-75P-302	Animal Physiology & Biochemistry Practicals based on Animal Physiology & Biochemistry			7	6
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
7	MJR	4	2	6	Yes	Lectures: 60 lectures including diagnostic and informative assessments during lecture hours and 30 Hours of Practical training/demonstration.
List of Programme Codes in which Offered as Minor Discipline		B.Sc. Chemistry: UG0804 B.Sc. Botany: UG0805				
Prerequisites		IV Semester (B.Sc. Bio group)				
Objectives of the Course:		The learning objectives of this course are as follows: ● To achieve a thorough understanding of various aspects of physiological systems and their functioning in animals. ● To expand their knowledge with respect to functioning of various organ systems such as muscular, nervous, digestive, circulatory, respiratory, excretory, reproductive and endocrine in animals. ● To instill the concept of hormonal regulation of physiology, metabolism and reproduction in animals. ● To provide insightful knowledge on the structure and classification of carbohydrates, proteins, lipids and enzymes.				

Detailed Syllabus

ZOO-75T-301 - Animal Physiology & Biochemistry

Unit - I

Physiology of digestion: Definition of digestion; various types of digestive enzymes and their digestive action in the alimentary canal, absorption and assimilation of the carbohydrates, proteins, and lipids.

5 Hrs

Physiology of respiration: Definition of respiration; mechanism of breathing; exchange of gases; transportation of oxygen and carbon dioxide in blood; regulation of respiration, respiratory pigments.

5 Hrs

Physiology of blood circulation: Composition and functions of blood; mechanism of blood clotting; heartbeat; cardiac cycle; blood pressure; body temperature regulation.

5 Hrs

Unit -II

Physiology of excretion: Kinds of nitrogenous excretory end products; Functional architecture of nephron, mechanism of urine formation, hormonal regulation of water and electrolyte balance (Homeostasis).

7

Hrs

Physiology of nerve impulse: Functional architecture of a neuron, origin and propagation of nerve impulse, synaptic transmission, reflex arc.

4 Hrs

Physiology of muscle contraction: Functional architecture of skeletal muscles; chemical and biophysical events during contraction and relaxation of muscle fibers.

4Hrs

Unit -III

Physiology of reproduction: Control and regulations of testis and ovarian functions; female reproductive cycles, implantation, parturition, and lactation in mammals. Testes, and ovaries.

6 Hrs

Physiology of endocrine glands: type of endocrine glands and their secretions and functions: Pituitary, adrenal, thyroid, pancreas.

6 Hrs

Hormonal regulation: mechanism of hormone action, feedback mechanism, hypothalamic control of pituitary function.

3 Hrs

Unit-IV

Carbohydrates: Structure, classification and function of carbohydrates, Glycolysis, Krebs's cycle, and oxidative phosphorylation and gluconeogenesis.

6 Hrs

Proteins: Structure, classification and function of proteins, deamination, transamination, decarboxylation, ornithine cycle, Enzymes; definition, classification, inhibition, and regulation.

5 Hrs

Lipids: Structure, classification and functions of lipids; Beta-oxidation; brief account of

biosynthesis of triglycerides.

4 Hrs

Suggested Books and References –

1. A TextBook of Medical Physiology (2017) Arthur C. Guyton MD, Eleventh ed., JohnE. Hall, Harcourt Asia Ltd.
2. Animal Physiology, Roger Eckert and Randal, 4th ed, Freeman Co, New York.
3. Text Book of Animal Physiology, Veer Bal Rastogi, Kedarnath Ramnath, Meerut.
4. Animal Physiology, Eckert R., Randall D. J., Burggen W., French K., 1997, Burggren WW & Co. Ltd.
5. Practical Zoology, Lal SS, 2014, Rastogi Publications.
6. Lehninger Principles of Biochemistry, Nelson DL and Cox MM, 2013, 6th edition W. H. Freeman.
7. Animal Physiology and Biochemistry, Sastry KV. 2nd edition Rastogi Publications.
8. Animal Physiology, Verma PS, Tyagi BS, Agarwal VK, 2004, 6th edition S. Chand & Co.
9. Biochemistry of Biomolecules, Ritu Kamal 2006, Paragon International Publishers
10. Biochemistry, Voet D and Voet JG, 2011, 4th edition, John Wiley & Sons, Inc.
11. Concepts of Biochemistry, Gupta S.N., Rastogi Publications Meerut.
12. Zoology for Degree Students, Agarwal VK, S Chand and Company New Delhi.
13. Principles of Anatomy and Physiology, Tortora, G.J. and Derrickson, B.H., 2009, XII Edition, John Wiley & Sons, Inc.

Suggested E-resources:

1. National Institute of Science Communication & Information Resources (NISCAIR) <http://nsdl.niscair.res.in/>
2. National digital library of India (NDL. India) <http://ndl.iitkgp.ac.in/>
3. https://onlinecourses.swayam2.ac.in/cec19_bt02/preview
4. https://onlinecourses.nptel.ac.in/noc20_bt42/preview (Animal Physiology)
5. Virtual Labs (<http://www.vlab.co.in>)
6. <http://www.ignouhelp.in/ignou-lse-05-study-material/> Animal Physiology (English Hindi) <http://rastogipublications.com/Rastogi-Publications-SUCHIPATRA-2022-23-Mail.pdf>
7. <http://rastogipublications.com/Rastogi-Publications-SUCHIPATRA-2022-23-Mail.pdf>
8. <https://www.schandpublishing.com/books/higher-education /sciences /zoology-degree-students-semester-iv/9789352534104/>
9. <https://www.scribd.com/document/696830782/A-K-Jain-Physiology-Practical-Manual>

Course Learning Outcomes:

By studying this course, students will be able to

- Have an enhanced knowledge and appreciation of animal physiology.
- Recognize and identify principal tissue structures and functions.

- Better understand the functions of important physiological systems including the nervous system, muscular system, endocrine and reproductive system.
- Understand how these separate systems interact to yield integrated physiological responses to maintain homeostasis in the body along with feedback mechanisms.
- Develop a strong foundation for research & employability skills using the knowledge of biochemistry.
- Improve their perspective of health through deep study of physiology and biochemistry.

Practical Syllabus

UG0802 -ZOO-75P-302- Practicals based on Animal Physiology & Biochemistry

1. Counting of RBCs and WBCs using haemocytometer.
2. Estimation of hemoglobin by Sahli's haemometer.
3. Estimation of haematocrit value (PCV).
4. Preparation of blood smear to study different blood cell types using Wright's or Leishman stain.
5. Demonstration of enzyme activity (catalase) in the liver.
6. Study the effect of pH and temperature on salivary amylase activity.
7. Study of histological sections of various organs and major endocrine glands: stomach, intestine, lungs, spleen, liver, kidney, spinal cord, pituitary, parathyroid, thyroid, pancreas, adrenal gland, testis, and ovary.
8. Detection of proteins in the animal tissue/food samples.
9. Detection of carbohydrates in the animal tissue/food samples.
10. Detection of lipids in the animal tissue/food samples.
11. Demonstration of separation of amino acids from a mixture using paper chromatography/TLC.

Scheme of Practical Examination and Distribution of Marks

S.No.	Practical Exercises	Regular	Ex. /N.C.
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R. J. Jais
 Dy. Registrar
 (Academic)
 University of Rajasthan
 JAIPUR

		Students	Students
1.	Major exercise (Physiology)	8	12
2.	Minor exercise (Physiology)	7	6
3.	Minor exercise (Biochemistry)	7	6
4.	Identification and comments on Spots (1 to 8)	16	16
5.	Viva Voce	7	10
6.	Class Record and report	5	
		50	50

Note:

***Internal marks for regular students only.**

1. With reference to microscopic slides, in case of non-availability, the exercise should be substituted with diagrams / photographs.
2. Candidates must keep a record of all work done in the practical class and submit the same for inspection at the time of the practical examination.
3. Mounting material for permanent preparations would be as per the syllabus or as available through collection and culture methods.
4. Computer aided techniques should be adopted as per UGC guidelines.